

## Practical tips for image acquisition

1. **Coverslips:** for fluorescence imaging use #1.5 coverslips
2. If performing **RNA FISH** make sure to treat with **Rnase Zap** all your accessories and to add **Rnase inhibitors** to your solutions and buffers.
3. **Minimizing light exposure:** When handling many coverslips simultaneously, add fluorescently labeled dyes to a small subset at a time to reduce light exposure. For example, when adding the hybridization mix containing dye-labeled oligos, apply it first to **2–4 coverslips**, place them at **30 °C in the dark**, and then proceed with the remaining coverslips.
4. For **DAPI** staining stay in the range of 20-100 ng/ml in PBS for 5' at RT
5. For **Hoechst** staining use 0.1ng/ul for a 30min-long staining and 5-10ng/ul for a 5min-staining
6. **Mounting media and imaging timing:** the preferred mounting medium is **GLOX buffer**. Ideally, image samples on the **same day or the following day** using GLOX (after sealing the coverslip with **Fixogum**). Afterwards, if needed, replace the mounting medium with **ProLong Gold ProLong Diamond** (if using far red) for long-term storage at **–20 °C**.
7. **Mounting with ProLong Gold:** after mounting in ProLong Gold, leave samples **in the dark at RT**. Transfer them to **–20 °C the following day (do not seal the coverslip)**. Ensure that excess ProLong Gold is removed using filter paper or gentle suction, as excess medium may cause imaging artifacts.
8. **Alternative mounting media (to GLOX):** if you prefer a non-solidifying medium and photobleaching is not a concern, **SlowFade** is a suitable alternative. It allows imaging on the same day, similar to GLOX.
9. **Oil removal before storage:** after imaging with immersion oil, remove the oil using **99% Isopropanol (IPA)**, followed by **70% ethanol**, before storing slides.
10. **Slide geometry:** ensure that the edges of the microscope slide are evenly supported and that the coverslip does not protrude beyond one edge. This prevents uneven seating of the slide in the microscope holder.
11. Seal your coverslips with **fixogum (always with non-solidfying media)**, not nail polish
12. **Chromatic aberration and channel alignment:** Even with “planar” objectives, differences in magnification and higher-order optical effects exist between

channels. Channel misalignment is common due to unavoidable minor misalignments of mirrors and filters. Unless there is a strong reason not to, use **TetraSpeck beads** or **multicolor fiducial markers** to correct chromatic aberration. This calibration must be performed **for every acquisition round**, on the **same day** as imaging. Daily temperature fluctuations can affect alignment, and the correction is specific to the **exact acquisition setup and individual imaging run**.

13. **Turn on** the Ti2 electronics, camera, and illumination at least **20–30 minutes** before imaging to allow thermal stabilization, especially for high-precision or multi-hour runs.
14. Acquire channels from the **longest to the shortest wavelength** (e.g. 700 nm → 375 nm) to minimize wavelength-dependent focus shifts during z-stack acquisition.
15. **Field-of-view selection**: image fields of view as randomly as possible, avoiding cherry-picking. Use the **random scanning option in JOBS** when available.
16. **Channel sequencing for non-colocalization studies**: if high-precision co-localization is not required, acquire **all z-planes for one channel first**, then switch channels and re-acquire the full stack. **Avoid changing filter sets at every focal plane**, as this greatly slows acquisition and may introduce unnecessary alignment errors. This can be configured in the **ND Acquisition panel** under *Order of Experiments* (Lambda/z-stack).
17. **Shutter control and photobleaching prevention**: ensure that all optical configurations have the **Spectra Shutter** option enabled so illumination is turned off when imaging is complete. Also enable shutter closure when moving between XY positions, switching lambda channels, and moving between focal planes. These options are set in **ND Acquisition** (see BICROscope documentation).
18. **Z-stack thickness**: acquire sufficiently thick z-stacks, typically **50–100 planes**. A stack is adequate when at least **5–10 empty planes** are present above and below the region containing signal. Insufficient z-coverage risks signal loss due to variable nuclear thickness and imperfect focus maintenance.
19. **Z-plane spacing**: proper spacing between z-planes is crucial. Steps that are too small increase photobleaching and acquisition time, while steps that are too large may miss signals or reduce z-localization accuracy. Optimal spacing depends on the **numerical aperture (NA)** of the objective. For a 100× objective with **NA = 1.45**, a step size of approximately **250 nm** is recommended. If uncertain, consult Erik or use an online calculator.
20. **Optimizing dynamic range**: to achieve a broad dynamic range, select a field of view containing both strong and weak signals. Adjust illumination intensity and gain

accordingly. Although 16-bit images can reach values up to **65,535**, the effective dynamic range may be limited by the sensor well depth, which depends on gain and other settings (refer to the camera datasheet). After setting parameters, check multiple fields of view (FOVs) to ensure robustness. Avoid using very bright, non-biological objects (e.g. debris) to set illumination. Exceeding well depth causes artifacts (e.g. streaking or flat saturated regions) that compromise downstream analysis; if observed, reduce exposure. To enhance weak signals increase the time of exposure (i.e. up to 2-3 s) while maintaining a broad effective dynamic range.

21. **Comparative imaging across samples:** when comparing signal intensities between samples, acquire all samples using **identical illumination and acquisition settings**. Set parameters using the **brightest sample first** to avoid saturation. Some samples may appear dim under these conditions. To analyze signal patterns within each sample independently, perform an additional acquisition with **sample-specific optimized settings**. These datasets should not be used for quantitative intensity comparisons, but are suitable for pattern analysis